RIDHOFIR FIKRI SHODIQ

Junior Telecommunications & Network Engineer

NOC · NETWORKING · NETWORK SECURITY · TELECOMMUNICATIONS

Associate Degree (D3) in Telecommunications Engineering, Politeknik Negeri Malang. Hands-on across BTS site documentation, NOC operations and network-security labs.

LOCATION
Sidoarjo, Jawa Timur, Indonesia

EMAIL
fikrishodiqridhofir@gmail.com

LINKEDIN
linkedin.com/in/ridhofir-fikri-shodiq

GITHUB
github.com/ridhofri



SIDOARJO, ID



FIG. 01 — POTRAIT

02 — PROFILE

Profile

Telecommunications Engineering graduate focused on network operations, telecommunications infrastructure and network security.

I completed internships in NOC operations at PT Gerbang Nusantara Sakti, monitoring 50+ network devices with PRTG, and in quality control and BTS site drafting at PT Poca Jaringan Solusi, where site drawings for a Telkomsel project were checked by Huawei and approved by Telkomsel.

I also build and document network-security labs: a BGP hijack detection lab using containerlab and FRRouting, and a Docker-based GSM/SS7 signaling lab using Osmocom. Both are isolated, reproducible environments published on GitHub.

CAREER DIRECTION

Junior roles in network engineering, NOC operations, RF / wireless and network security.

EDUCATION

Associate Degree (D3) in Telecommunications Engineering
Politeknik Negeri Malang

LOCATION

Sidoarjo, East Java
Indonesia

FOCUS

Network · NOC · Network Security · Radio Frequency · FTTx · Internet of Things / Wireless Sensor Network

CERTIFICATIONS

MTCNA · CCNA · BNSP · CSCU · Fiber Optic

03 — EDUCATION

Education

Formal education in telecommunications — from a vocational access-network programme to an associate degree (D3).

2022 – 2025
ASSOCIATE · D3

E-01 · HIGHER EDUCATION

Telecommunications Engineering

GPA
3.44 / 4.00

DEGREE
Associate Degree (D3)

INSTITUTION
Politeknik Negeri Malang

PERIOD
Aug 2022 – Jul 2025

2019 – 2022
VOCATIONAL · SMK

E-02 · VOCATIONAL SECONDARY EDUCATION

Telecommunications Access Network Engineering

LEVEL
SMK — Vocational high school

INSTITUTION
SMK Telkom Sidoarjo

PERIOD
Jul 2019 – Jul 2022

04 — TECHNICAL SKILLS

Technical skills

Grouped by domain. No proficiency bars — the evidence sits on the project and certification pages.

S-01 · 11 ITEMS

Networking

- TCP/IP
- IPv4/IPv6
- Subnetting
- VLAN
- Routing
- Switching
- OSPF
- BGP
- DHCP
- DNS
- NAT

S-02 · 08 ITEMS

Telecommunications

- GSM / LTE
- BTS
- RF Planning
- Link Budget
- COST-231 Hata
- FTTH / FTTx
- Splicing
- OTDR / OPM

S-03 · 04 ITEMS

Monitoring

- PRTG
- Ping
- Traceroute
- Wireshark

S-04 · 03 ITEMS

Devices

- MikroTik
- Cisco
- Huawei

S-05 · 07 ITEMS

Tools

- AutoCAD
- Packet Tracer
- Winbox
- Linux
- Atoll
- Arduino IDE
- Excel

S-06 · 05 ITEMS

Wireless / IoT

- ESP32
- nRF24L01
- LoRa / LoRaWAN
- MQTT
- WSN

S-07 · 03 ITEMS

Security

- Network Security Fundamentals
- BGP Hijacking (lab)
- Telecommunications Security

05 — EXPERIENCE

Experience

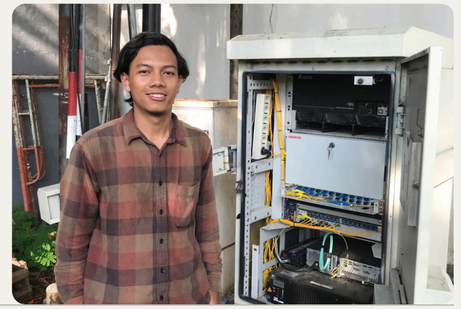
Internships and early roles — network operations, BTS infrastructure and telecom administration.

Jan – Feb 2026
INTERNSHIP

PT Gerbang Nusantara Sakti

NETWORK OPERATIONS CENTER (NOC)

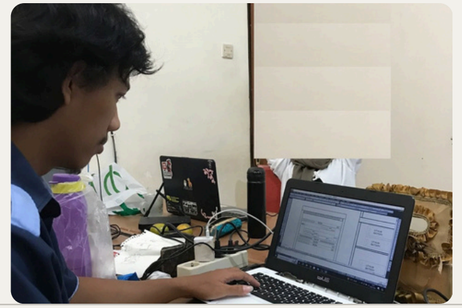




Jul – Oct 2024
INTERNSHIP

PT Poca Jaringan Solusi

QUALITY CONTROL





Jul – Oct 2024
INTERNSHIP

PT Poca Jaringan Solusi

DRAFTER

Jun – Nov 2021
INTERNSHIP

PT Telkom Akses Sidoarjo

WAREHOUSE ASSISTANT





Jan – Feb 2020

PT Telkom Akses Sidoarjo

TEACHING FACTORY - DATA ENTRY

06 — FEATURED PROJECTS

Featured projects

Ten projects across research, lab, internship and final-project work. Page numbers point to each case study.

01 BGP Hijack Detection Lab

INDEPENDENT PROJECT

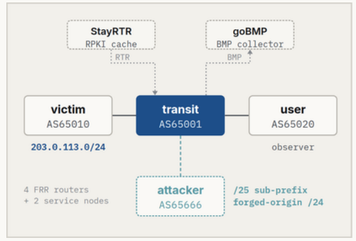


FIG. 01.1 — LAB TOPOLOGY

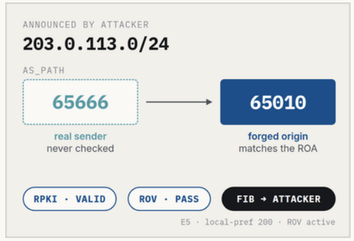


FIG. 01.2 — FORGED-ORIGIN BYPASS

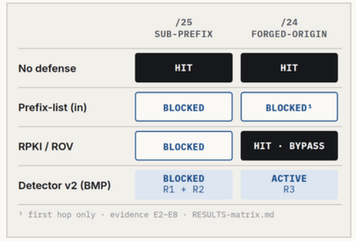


FIG. 01.3 — ATTACK × DEFENSE RESULTS

OBJECTIVE

Reproduce BGP prefix hijacks in an isolated container lab, measure standard defenses with before/after tests, and detect where they break.

BUILD

Four FRR routers on private ASNs, a StayRTR RPKI cache and a goBMP collector, deployed from one containerlab file. Attacks and restores are scripted; detectors are written in Python.

KEY FINDING

A forged-origin hijack (AS_PATH 65666 65010) is marked RPKI-valid and slips past ROV. An ASPA-style invalid-adjacency rule (R3) catches it; BMP views label each route BLOCKED, ACTIVE or WINNING.

SCOPE

Simulation only: RFC 5737 prefixes, private ASNs, no public peering. Nine experiments logged (E1–E9), Sep 2026.

containerlab · FRR 10.7.1 · StayRTR · goBMP · Python 3

github.com/ridhofri/bgp-hijack-detection-lab

02 GSM 2G & SS7/SIGTRAN Security Lab

ACADEMIC PROJECT

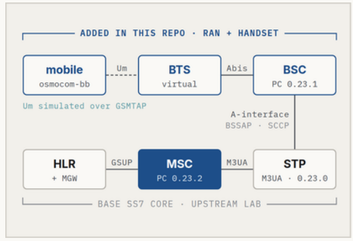


FIG. 02.1 — LAB ARCHITECTURE

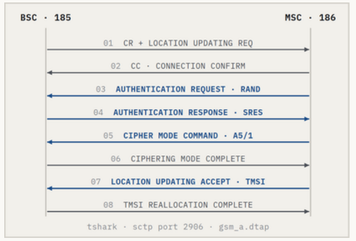


FIG. 02.2 — LOCATION UPDATE · DTAP

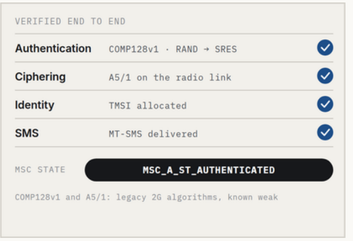


FIG. 02.3 — VERIFIED PROCEDURES

OBJECTIVE

Turn a core-only SS7 signaling lab into a complete 2G network and trace each subscriber-security step in real protocol captures.

BUILD

Extended Tagurkrishna/SS7-Simulation-Lab with OsmoBSC, a virtual BTS and an osmocom-bb handset built from source. Everything runs in Docker; the radio is simulated over GSMTAP, so nothing is transmitted.

RESULT

The A-interface capture shows a full authenticated Location Update: COMP128v1 challenge, A5/1 ciphering, TMSI reallocation. MSC state MSC_A_ST_AUTHENTICATED; MT-SMS delivered.

TROUBLESHOOTING

Fixed VTY port clashes between handset containers and an SMS failure (RP cause 22) caused by a missing sms.txt. Documented in a README and a 12-section report.

Osmocom · Docker Compose · osmocom-bb · Wireshark

github.com/ridhofri/ss7-gsm-lab-extended

PROJECTS 03 – 04 — RF · TELECOMMUNICATIONS

More Projects

03 Signalink — Link Budget Calculator

INTERNSHIP PROJECT · TEAM



FIG. 03.1 — MENU



FIG. 03.2 — AREA TYPE

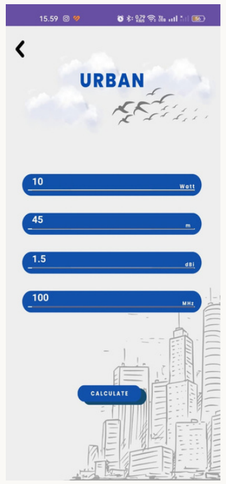


FIG. 03.3 — PARAMETERS



FIG. 03.4 — RESULT

ROLE

Team project during the PT Poca Jaringan Solusi internship, Jul – Oct 2024. My part: link-budget parameters, app testing and checking results against theory.

MODEL

COST-231 Hata for urban and suburban areas. Inputs: Tx power, antenna gain and height, frequency. Outputs: coverage distance (D) and path loss.

Android app · COST-231 Hata · team

Quick link-budget estimates for technicians

04 BTS Site Survey Drawings

INTERNSHIP PROJECT · 50+ DRAWINGS · 20+ SITES

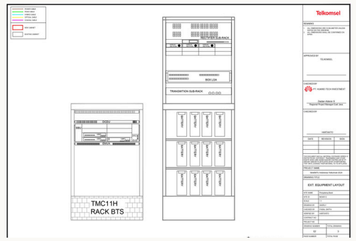


FIG. 04.1 — EQUIPMENT LAYOUT

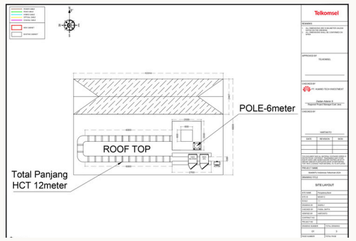


FIG. 04.2 — ROOFTOP SITE LAYOUT

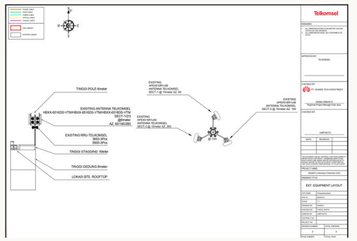


FIG. 04.3 — ANTENNA LAYOUT

01	02	03	04	05
Field survey	**Survey data**	**AutoCAD drawing**	**Vendor check**	**Operator approval**
survey team	cables · batteries · azimuths	drafted by me	Huawei	Telkomsel

ROLE

Drafter intern at PT Poca Jaringan Solusi, Jul – Oct 2024. Turned field-survey data into installation-ready AutoCAD drawings for Telkomsel BTS sites: 50+ drawings across 20+ sites.

SAMPLE DRAWINGS

Equipment layout (TMC11H BTS rack, rectifier and battery sub-racks), a rooftop site with a 6 m pole and 12 m HCT cable run, and sector antennas with azimuths and RRU positions.

AutoCAD · site survey · As-Built Documentation

Checked by Huawei · approved by Telkomsel

PROJECTS 05 – 06 — IOT · COMPUTER VISION

More Projects

05 Smart Greenhouse Irrigation System IoT-Based Adaptive Irrigation with Fuzzy Tsukamoto

PROJECT CONTRIBUTION · IOT



FIG. 05.1 — WATER RESERVOIR



FIG. 05.2 — IRRIGATION LINE

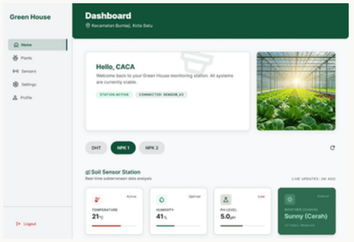


FIG. 05.3 — WEB DASHBOARD

OBJECTIVE

Replace fixed manual watering of kale in a greenhouse with an IoT system that sets watering time from soil data and the weather forecast.

BUILD

ESP32 reads an RS485 soil sensor and switches a pump and solenoid valve through a relay module. A Laravel backend combines soil data with a weather-forecast API and runs Fuzzy Tsukamoto (16 rules) to set watering time.

RESULTS

Fuzzy output matched hand calculations in all test cases. On one bed over eight scheduled waterings (17–21 Jul 2026), the system used 48.4 L versus 72 L for manual watering, about 33% less.

MY CONTRIBUTION

Helped troubleshoot the hardware so the system waters automatically at the scheduled times (07:00 and 15:00).

ESP32 · RS485 soil sensor · Laravel · MQTT · Fuzzy Tsukamoto

Informatics student's thesis · Polinema · 2026

06 ESP32-CAM + YOLOv8 Pose

ACADEMIC PROJECT · COMPUTER VISION / IOT



FIG. 06.1 — ESP32-CAM ENCLOSURE

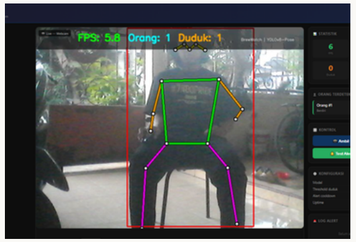


FIG. 06.2 — POSE DETECTION



FIG. 06.3 — TELEGRAM ALERT

OBJECTIVE

Monitor employee activity with a low-cost camera: detect people and their posture in real time and alert on prolonged sitting.

ARCHITECTURE

An ESP32-CAM in Access Point mode streams to a Python Flask server running YOLOv8-Pose. Alerts go to a web dashboard, a Telegram bot and an active buzzer.

TESTING

Detection tested from front, side and back angles at several distances, measured by FPS and detection success rate.

DEBUGGING

Fixed an ESP32-CAM camera-init failure by selecting the correct board profile and enabling PSRAM. The demo alert fires after 10 s of sitting.

ESP32-CAM · Flask · YOLOv8-Pose · Telegram Bot

PROJECTS 07 – 08 — WIRELESS · IOT · EMBEDDED

More Projects

07 Meshtastic ESP32 — LoRa Mesh

INDEPENDENT PROJECT - LORA / WIRELESS NETWORKING



FIG. 07.1 — TWO LORA NODES



FIG. 07.2 — NODE MAP



FIG. 07.3 — NODE-TO-NODE LINK

OBJECTIVE

Build Meshtastic LoRa mesh nodes from generic parts, from wiring to firmware, instead of using a ready-made board.

BUILD

Two nodes, each an ESP32 wired over SPI to an Ai-Thinker Ra-02 (SX1278) LoRa module, in a plastic enclosure with an external antenna. Flashed Meshtastic (DIY_V1 target) and set the role, channel and region.

RESULT

Node RAPTOR_1 came online and appeared on the public Meshtastic map through its own MQTT uplink. Planned distance between the two sites: 891 m, measured in Google Earth.

SCOPE

Solo build: I designed and assembled both nodes, from wiring and enclosure to firmware setup.

ESP32 · Ai-Thinker Ra-02 (SX1278) · Meshtastic · MQTT

Independent project · 2026

08 Biogas Monitoring System

IOT / EMBEDDED SYSTEM

D3 final project (team) · Politeknik Negeri Malang · Jan – Jun 2025



FIG. 08.1 — REACTOR PROTOTYPE



FIG. 08.2 — LCD READOUT

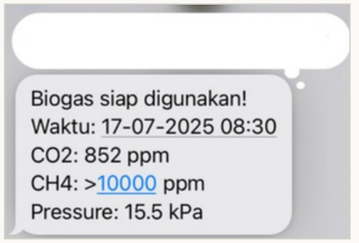


FIG. 08.3 — SMS ALERT

OBJECTIVE

Produce biogas from cow manure and monitor gas concentration, pressure and temperature in real time.

IMPLEMENTATION

Sensors on the reactor feed live readings to an on-site LCD and flag abnormal values; an SMS alert is sent when the biogas is ready.

SAMPLE READINGS CAPTURED ON THE DEVICE

```
LCD · 06-07-2025   CO₂ 784 ppm · CH₄ 888 ppm · 27.2 °C · 1.8 kPa
SMS · 17-07-2025   CO₂ 852 ppm · CH₄ >10000 ppm · 15.5 kPa
```

Role: [PLACEHOLDER: your part — firmware, sensors, wiring, reactor]








PROJECTS 09 – 10 — IOT · RF PLANNING

More Projects

09 Smart Street Light

ACADEMIC PROJECT · IOT

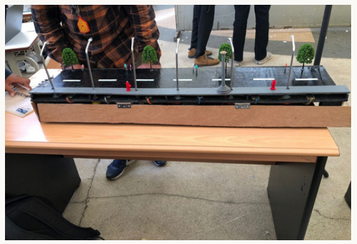


FIG. 09.1 — PROTOTYPE DEMO

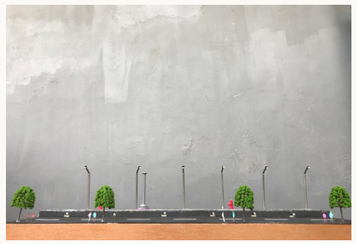


FIG. 09.2 — STREET MODEL



FIG. 09.3 — LAMP POLE DETAIL

OBJECTIVE

Cut street-light energy waste: lamps stay off by day, dim to standby at night and switch to full brightness only where a vehicle or pedestrian passes.

BUILD

ESP32 DevKit V1 reads an LDR and six IR obstacle sensors and drives six LED street lights on a scale road model; each IR sensor controls its own lamp. Blynk Cloud over WiFi adds remote monitoring and manual control.

RESULT

Working road model: after two faulty IR modules and two dim LEDs were replaced, each lamp followed its own sensor as programmed. IR detection reached about 15 cm against a 20 cm target.

MY ROLE

Product Owner in an eight-person Scrum team. Soldered the ESP32 headers, resistors and connectors onto perfboard, built the IR sensor mounts and cables, and 3D-printed the LDR housing.

ESP32 DevKit V1 · LDR · 6× IR sensors · Blynk

Project Design course · team of 8 · Polinema · 2024

10 LTE RF Planning & Atoll Simulation

ACADEMIC PROJECT · RF PLANNING / LTE

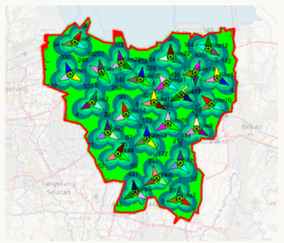


FIG. 10.1 — 24-SITE PLAN · JAKARTA

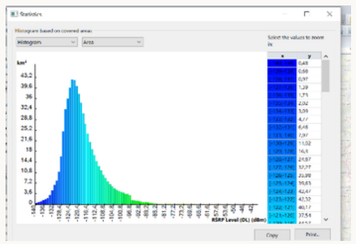


FIG. 10.2 — RSRP PREDICTION (DL)

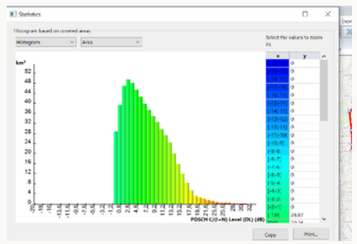


FIG. 10.3 — PDSCH C/(I+N) (DL)

OBJECTIVE

Plan an LTE network for dense-urban Jakarta: size the number of sites from capacity and from coverage, then check the coverage-based plan with an Atoll prediction and simulation.

METHOD

Capacity: a population forecast from BPS data, service and traffic models and site throughput gave 11 sites. Coverage: a downlink link budget and COST-231 Hata path loss gave 24 sites; this larger plan was simulated.

RESULTS

Atoll simulation: 41,540 of 47,188 users connected (88%). Predicted mean RSRP −119.31 dBm and mean PDSCH C/(I+N) 6.75 dB; cumulated application throughput 256.49 Mbps DL and 342.17 Mbps UL.

KEY FINDING

The 5,648 rejected users (12%) were signal gaps, not capacity: 1,016 no coverage, 4,632 no service, zero saturation. Next step: optimise site placement and power control.

Atoll · LTE · link budget · COST-231 Hata · BPS data

Cellular Communication Systems lab · individual · Polinema · 2024

11 — CERTIFICATIONS

Certifications

Details transcribed from the original certificates. Course completion is labelled as a course, not a certification.



CERTIFICATION

MikroTik Certified Network Associate (MTCNA)

ISSUER
MikroTik

DATE
03 Aug 2026

ID / NOTE
2608NA2483



COMPETENCY · BNSP

Access Network Planner (Perencana Jaringan Akses)

ISSUER
BNSP · Telecommunication Engineering

DATE
08 Sep 2025 · valid 3 yrs

ID / NOTE
43212.3522.00.5.0012882.2025



CERTIFICATION

Certified Secure Computer User (CSCU)

ISSUER
EC-Council

DATE
11 Sep 2026

ID / NOTE
ECC4975026831



COURSE COMPLETION

CCNAv7: Introduction to Networks

ISSUER
Cisco Networking Academy

DATE
20 Jan 2024

ID / NOTE
Course — not the CCNA exam



COMPETENCY ASSESSMENT

Fiber Optic Installation

ISSUER
SMK Telkom Sidoarjo · PT Fangbian Iskan Corporindo

DATE
20 Apr 2022

ID / NOTE
Result: Very Competent

12 — ACHIEVEMENTS

Achievements

Competition results, verified from the certificates.







GOLD

2024 · GOLD MEDAL

Mandalika Essay Competition 4

Faculty of Agriculture, Universitas Nahdlatul Wathan Mataram, with Lembaga Setara Prisma Nusantara · Lombok, 2–4 March 2024

Optimization of Inventory and Sales Monitoring Systems Through Mitragro to Stabilize Prices and Promote Sustainable Agriculture and Livestock in Malang Regency



SILVER

2024 · SILVER MEDAL

EL BAJO Essay Competition 3

National essay competition by Dinas Pendidikan, Kepemudaan dan Olahraga Kab. Manggarai Barat and Sentosa Foundation · Labuan Bajo, 6–7 October 2024

SMOB (Smart Eco Biogas) Based on IoT: Transforming Organic Waste into Renewable Energy and Compost



3RD

2023 · 3RD PLACE

LoGraK 2023

Lomba Gagasan dan Rancangan Kreatif Nasional (national creative ideas & design competition) · Renewable Energy category · Electrical Engineering Department, Politeknik Negeri Malang · 23 November 2023

Transforming Organic Waste into Renewable Energy and Compost through IoT-Based SMOB (Smart Eco Biogas): An Innovative Solution for Sustainable Renewable Energy Development and Achieving Sustainable Development Goals 7 and 9



13 — ORGANIZATIONS

Organizations

Two organizational roles, Aug 2022 – Jul 2025 — public relations and information & technology.

O-01

LEADERSHIP · COMMUNICATION

Forum Mahasiswa Sidoarjo

Head of Public Relations Division | Aug 2022 – Jul 2025

CONTRIBUTION

- Managed external relations with 10+ university and student-organization representatives.
- Developed and distributed organizational information via communication channels and social media, increasing member awareness.
- Built relationships with 100+ members, alumni and community stakeholders.





O-02

TEAMWORK · TECHNICAL SUPPORT

Workshop Electro Polinema

Member — Information & Technology | Aug 2022 – Jul 2025

CONTRIBUTION

- Troubleshot devices and network connectivity at 15+ organizational events.
- Managed 30+ laptops, projectors and network devices for event operations.
- Installed, configured, and tested network infrastructure for 10+ events, ensuring reliable connectivity and optimal network performance.





14 — CONTACT

Let's connect.

Open to junior roles in network engineering, NOC operations, RF / wireless and network security.

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GITHUB	github.com/ridhofri
WHATSAPP	+62 821-3150-7876
LOCATION	Sidoarjo, East Java, Indonesia

WHATSAPP

EMAIL

LINKEDIN

GITHUB



NETWORK MAP — ILLUSTRATIVE